Substituting an expression

Replace a variable with a linear expression, then expand and simplify.

Name: _____ Suggested time: 10 minutes

Show your working. Substitute carefully and simplify.

01 Given x=3y-2

Write in terms of y and simplify

Simplify: 6x-7

02 Given x=6y+5

Write in terms of y and simplify

Simplify: 3x+3

03 Given x=2y+1

Write in terms of y and simplify

Simplify: 5x-2

04 Given x=4y+4

Write in terms of y and simplify

Simplify: 4x-7

Answer sheet

Substituting an expression | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $18y-19$

02 $18y+18$

03 $10y+3$

04 $16y+9$

Worked solutions

Substituting an expression

01 Given x=3y-2

6x-7

Substitute x=3y-2.

6(3y-2) -7

18y -12 -7

18y-19

02 Given x=6y+5

3x+3

Substitute x=6y+5.

3(6y+5) +3

18y +15 +3

18y+18

03 Given x=2y+1

5x-2

Substitute x=2y+1.

5(2y+1) -2

10y +5 -2

10y+3

04 Given x=4y+4

4x-7

Substitute x=4y+4.

4(4y+4) -7

16y +16 -7

16y+9